UNIVERSITY OF MAURITIUS
SIS 3204Y(5) – NATURAL LANGUAGE PROCESSING

MOCK PAPER 5 – Weeks 1 to 8 (Question Paper with Model Answers)
Total: 50 marks | Suggested duration: 60 minutes

Instructions: Answer ALL questions. Marks are shown beside each question. The week each question comes from is shown in the question header. Model answers are given under each question in bullet points; the mark for a point is shown in brackets, e.g. (1). Cover the answers while practising. Equivalent correct answers should receive credit.

Question 1 | Week 1 (Revision Q11) [5 marks]

Explain Retrieval-Augmented Generation (RAG) and discuss its purpose.

Answer

- **Concept:** RAG combines **external information retrieval** with a generative language model. (1)
- **Process:** factual documents are fetched from an external source and given to the generator as context. (1)
- **Purpose:** answers are grounded in verifiable, domain-specific or up-to-date data. (1)
- It **reduces hallucinations**. (1)
- It increases the **reliability** of the output. (1)

Question 2 | Week 2 (Revision Q17) [4 marks]

Explain what sentence tokenisation is and describe FOUR challenges that make it difficult.

Answer

- **Sentence tokenisation** splits a paragraph or document into individual sentences; it helps systems understand structure and boundaries. (1)
- Challenges (1 mark each, any three):
- **Abbreviations** (Dr., Mr., U.S.A.) can cause wrong splits.
- **Exclamations and questions** make sentence ends less clear.
- **Ellipses (...) and multiple punctuation** marks.
- **New lines** are not always real sentence breaks.
- **Domain-specific writing styles** follow different rules.

Question 3 | Week 3 (Revision Q15) [12 marks]

Trace the evolution of text representation from integer encoding to Word2Vec. Show how each technique tries to fix the weakness of the previous one and why Word2Vec is a step forward.

Answer

- **Starting point:** computers understand only numbers, so text goes through preprocessing, tokenisation, vocabulary building, indexing and encoding. (1)
- **Integer encoding:** each token gets a unique ID; simple, memory efficient, a good baseline. (1)
- **Weakness:** false order and distance (e.g. zebra=4 > apple=0), which misleads linear and distance-based models; no semantics; sensitive to vocabulary changes. (2)
- **One-hot fixes this:** a binary vector per word, so no order and no distance; all words are treated equally; works with linear models and neural networks. (2)
- **Weakness:** sparse, high-dimensional vectors (100,000 words → 99.999% zeros), high memory and computation; all words equidistant so no meaning; no frequency. (2)
- **BoW / TF-IDF:** fixed-length document vectors with frequency/importance, but still sparse, ignore word order and carry no meaning. (1)
- **Word embeddings fix this:** dense, low-dimensional (50–300) vectors where similar words are close and relationships appear (king − man + woman ≈ queen). (1)
- **Word2Vec:** learns embeddings automatically from raw text (distributional hypothesis) using CBOW or Skip-Gram; lower memory, faster, better generalisation. (1)
- **Remaining gap:** Word2Vec is static and local, and cannot handle polysemy or OOV words, which led to contextual embeddings (ELMo, BERT, GPT). (1)

Question 4 | Week 4 (Part B Q5) [3 marks]

Why can’t a neural network directly process “This movie is excellent”? How can embeddings be used?

Answer

- Neural networks work on **numerical values**, not raw text. (1)
- The sentence is **tokenised** and each word is mapped to an embedding. (1)
- The **dense embedding vectors** are given to the network as numerical input. (1)

Question 5 | Week 5 (Additional Q13) [8 marks]

Describe how an RNN can be used to classify a sentence into one of three categories: positive, neutral or negative.

Answer

- The sentence is **tokenised** and prepared. (1)
- Each token is represented using an **embedding**. (1)
- The embeddings are processed **one at a time** by the RNN. (1)
- At each time step the **hidden state** is updated using the current word embedding and the previous hidden state. (2)
- After the sequence, a representation such as the **final hidden state** is passed to an output layer. (1)
- A **softmax** layer gives probabilities for positive, neutral and negative. (1)
- The class with the **highest probability** is the prediction. (1)

Question 6 | Week 6 (Set 2 Q14) [9 marks]

A student says: “Q, K and V are identical because they all come from the same input sequence.” Explain why this statement is incorrect.

Answer

- The statement is **incorrect**. (1)
- Q, K and V may all come from the same input representations X. (1)
- But they are produced with **different learned projection matrices**. (2)
- $Q = XW_Q$, $K = XW_K$, $V = XW_V$. (2)
- **Query:** used to look for relevant information. (1)
- **Key:** compared with the Query to decide relevance. (1)
- **Value:** holds the information that is combined to make the output representation. (1)

Question 7 | Week 7 (Set 1 Q2) [5 marks]

a) Explain the purpose of multi-head self-attention in a Transformer encoder. (2 marks)

b) Describe the role of the position-wise feed-forward network. (2 marks)

c) State the purpose of layer normalisation. (1 mark)

Answer

(a)

- Multiple attention heads work **in parallel**. (1)
- Different heads can capture different **syntactic and semantic relationships**. (1)

(b)

- The FFN transforms **each token representation independently**. (1)
- It uses linear layers and a **non-linear activation** to enrich the representations. (1)

(c)

- Layer normalisation **normalises features** and **stabilises training**. (1)

Question 8 | Week 8 (Tutorial Set A Q2) [4 marks]

(a) Explain the role of pre-training in developing an LLM. (2 marks)

(b) Explain two limitations of LLMs. (2 marks)

Answer

(a)

- Pre-training exposes the model to **large amounts of text**. (1)
- It learns **statistical patterns and language representations**, often through next-token prediction. (1)

(b)

- **Hallucination:** LLMs can produce plausible but incorrect information. (1)
- **Bias:** LLMs can reproduce biases found in their training data. (1)

Marks Summary

Question	Week	Source	Marks
1	1	Revision Q11	5
2	2	Revision Q17	4
3	3	Revision Q15	12
4	4	Part B Q5	3
5	5	Additional Q13	8
6	6	Set 2 Q14	9
7	7	Set 1 Q2	5
8	8	Tutorial Set A Q2	4
		TOTAL	**50**